

Se Min Kong

AI | ROBOTICS | COMPUTER VISION

semin1224@gmail.com | +82 10-9179-0318 | github.com/SeMinKong | DOB: 24 Dec 2000

PROFILE

Software engineering graduate focused on AI and robotics, with hands-on experience integrating computer vision, ROS 2, backend services, and hardware. Builds end-to-end systems spanning model pipelines, real-time control, and user-facing dashboards.

EDUCATION

Soongsil University | College of IT, School of Software Mar 2020 - Feb 2026
Bachelor of Engineering in Software; Major in Artificial Intelligence & Big Data; Convergence Major in Big Data
GPA 3.61/4.50 | Graduated 20 Feb 2026
Dulwich College Suzhou (DCSZ) | Primary & Middle School, Suzhou, China 2008 - 2012

TECHNICAL SKILLS

Languages & Core: Python, C++, TypeScript
AI & Vision: PyTorch, Ultralytics YOLO, OpenCV, Transformers, CUDA
Robotics & Systems: ROS 2, Intel RealSense, Dobot Magician, RoboDK, NVIDIA Isaac Sim / Isaac Lab
Backend & Tools: FastAPI, WebSocket, React, Node.js, Socket.io, Git, Docker, Linux (Ubuntu)

EXPERIENCE

Samsung SW-AI Academy For Youth (SSAFY) | Trainee, Robotics Track Jan 2026 - Present

- Completing an intensive SW-AI curriculum centered on robotics, computer vision, ROS 2, and integrated hardware-software projects.

Republic of Korea Army | Tactical Communications Equipment Operator/Maintainer May 2021 - Nov 2022

- Completed mandatory military service and was discharged as a Sergeant.

SELECTED PROJECTS

AQIS for Smart Factory | Team Lead, Full-Stack & Robot Integration May 2026 - Jun 2026
ROS 2 | FastAPI | WebSocket | React | RealSense | Dobot | YOLOv5 | [Repository](#)

- Led full-stack and robot integration on a two-person team for a smart-factory system: RealSense + YOLOv5 inspection triggered conveyor stops and Dobot vacuum pick-and-place.
- Built the RealOps dashboard, FastAPI REST/WebSocket services, ROS 2 bridge, device adapters, and LLM command integration for live monitoring and control.

Briefit | AI-Based News Curation Service | Team AI Engineer May 2025 - Sep 2025
Python | aiohttp | BeautifulSoup | Transformers | KoBART | [Repository](#)

- Built an asynchronous multi-source news pipeline with similarity grouping and neutral article summarization.
- Worked on crawling, preprocessing, tokenization, and KoBART summarization experiments across API- and local-model pipeline variants.

Brain Tumor MRI Classification & Segmentation | Personal Project Feb 2026 - Apr 2026
Python | PyTorch | Ultralytics YOLO11 | OpenCV | NumPy | [Repository](#)

- Built a unified YOLO11 classification and instance-segmentation pipeline for four MRI classes, combining diagnosis and tumor-mask visualization in one output.
- Automated grayscale mask-to-polygon label conversion; reported 99.4% Top-1 accuracy and 92.7% Mask mAP50 in project evaluations.

Alkkagi.io | Personal Project Mar 2026 - Apr 2026
React 19 | TypeScript | Node.js | Express | Socket.io | [Repository](#)

- Built a real-time multiplayer browser game with a React/TypeScript frontend and Node.js/Express/Socket.io backend.
- Implemented a server-authoritative 60 FPS physics engine for elastic collisions, friction, momentum, position correction, and dynamic mass/radius scaling.

AWARDS & CERTIFICATIONS

2025 IT Project Pro League - Encouragement Award | Soongsil University Spartan SW Education Center 22 Nov 2025
Software Competition - Gold Prize | Soongsil University SW
Capstone Design Competition - Encouragement Award | Soongsil University SW
OPic English - IH (Intermediate High) | Tested 5 Oct 2025 | Valid through 4 Oct 2027